

Here's your **MCQ-based quiz (without answers)** for B.Tech students, covering all requested topics in a structured and practical way.

Azure Cognitive Services & AI Deployment Quiz (MCQs)

SECTION 1: Cognitive Services Overview & Architecture

Q1.

Which Azure service is best suited for extracting structured data from invoices and forms?

- A. Speech Service
 - B. Language Service
 - C. **Document Intelligence**
 - D. Computer Vision
-

Q2.

In Azure Cognitive Services architecture, what is the primary role of an endpoint?

- A. Store training data
 - B. **Provide access to AI models via API**
 - C. Manage user authentication
 - D. Monitor application logs
-

Q3.

Which component is responsible for converting speech into text?

- A. Translator
 - B. **Speech Service**
 - C. Vision Studio
 - D. Bot Framework
-

Q4.

Which architecture pattern is commonly used when multiple AI services are combined?

- A. Monolithic architecture
 - B. **Microservices architecture**
 - C. Peer-to-peer
 - D. Standalone system
-

Q5.

Which Azure service helps detect objects, faces, and text within images?

- A. Language Service
 - B. **Computer Vision**
 - C. Speech Service
 - D. Form Recognizer
-

SECTION 2: Service Categories & Use Cases

Q6.

Which service is used for sentiment analysis?

- A. Computer Vision
 - B. **Language Service**
 - C. Speech Service
 - D. Translator
-

Q7.

Which use case best fits the Speech Service?

- A. Image classification
 - B. **Real-time voice assistant**
 - C. Document parsing
 - D. Language translation
-

Q8.

Which service is most appropriate for building a multilingual chatbot?

- A. **Translator + Language Service**
 - B. Computer Vision
 - C. Face API
 - D. Form Recognizer
-

Q9.

Which category does OCR (Optical Character Recognition) belong to?

- A. Speech AI
 - B. **Vision AI**
 - C. Language AI
 - D. Decision AI
-

Q10.

Which scenario requires combining multiple Cognitive Services?

- A. Converting text to speech
 - B. Detecting faces in an image
 - C. **Analyzing video with audio and subtitles**
 - D. Translating text
-

SECTION 3: API Integration Patterns

Q11.

Which protocol is most commonly used to call Azure Cognitive Services?

- A. FTP
 - B. **REST**
 - C. SMTP
 - D. TCP
-

Q12.

What is required to authenticate API calls to Azure Cognitive Services?

- A. Username and password
- B. **API key or token**

- C. Database connection string
 - D. IP address
-

Q13.

Which integration method provides pre-built functions and easier coding?

- A. REST API
 - B. SDK
 - C. CLI
 - D. Portal
-

Q14.

Which pattern is best for handling large-scale asynchronous processing?

- A. Blocking calls
 - B. Batch processing with queues
 - C. Direct synchronous calls
 - D. Manual processing
-

Q15.

What is the purpose of retry logic in API integration?

- A. Improve UI design
 - B. Handle temporary failures
 - C. Reduce API cost
 - D. Increase response time
-

SECTION 4: Deployment Strategies

Q16.

Which deployment model is best for applications requiring global scalability?

- A. On-premises
- B. Cloud-based deployment
- C. Local server
- D. Standalone system

Q17.

Which deployment is ideal for low-latency and offline scenarios?

- A. Cloud deployment
 - B. **Edge deployment**
 - C. Hybrid deployment
 - D. SaaS deployment
-

Q18.

What is a key advantage of container-based deployment?

- A. Requires no internet
 - B. **Easy portability across environments**
 - C. No need for security
 - D. No configuration required
-

Q19.

Which Azure feature helps automatically scale applications?

- A. Virtual Machines
 - B. **Auto-scaling**
 - C. Storage Accounts
 - D. Resource Groups
-

Q20.

Which strategy helps reduce cost in AI deployments?

- A. Always using highest pricing tier
 - B. **Scaling resources based on demand**
 - C. Avoiding monitoring
 - D. Using only on-premises systems
-

SECTION 5: Identity & Access Management (IAM)

Q21.

Which service is used for identity management in Azure?

- A. Azure DevOps
 - B. Azure Active Directory
 - C. Azure Storage
 - D. Azure Monitor
-

Q22.

Which authentication method is more secure?

- A. API keys
 - B. Managed Identity
 - C. Password-based login
 - D. Anonymous access
-

Q23.

What is the main purpose of role-based access control (RBAC)?

- A. Improve UI
 - B. Control user permissions
 - C. Increase speed
 - D. Store data
-

Q24.

What should you do if an API key is exposed publicly?

- A. Ignore it
 - B. Regenerate the key immediately
 - C. Share it with team
 - D. Disable the application
-

◆ SECTION 6: Responsible AI, Security & Compliance

Q25.

Which principle ensures AI systems treat all users fairly?

- A. Reliability
- B. Fairness

- C. Transparency
 - D. Privacy
-

Q26.

Which technique helps reduce harmful AI outputs?

- A. Increasing model size
 - B. Guardrails and content filtering
 - C. Removing APIs
 - D. Disabling authentication
-

Q27.

What is the purpose of data encryption?

- A. Improve speed
 - B. Protect sensitive data
 - C. Reduce storage
 - D. Simplify UI
-

Q28.

Which regulation focuses on data privacy?

- A. HTTP
 - B. GDPR
 - C. TCP
 - D. HTML
-

Q29.

What is bias in AI systems?

- A. Faster processing
 - B. Unfair or skewed outcomes
 - C. High accuracy
 - D. Secure processing
-

Q30.

Which practice improves AI transparency?

- A. Hiding model logic
- B. Providing explanations for predictions
- C. Ignoring outputs
- D. Reducing data